NEXT ENGINEER · _SHEERAT_

# Essence of Indian Traditional Knowledge

**BNC602**

One-day revision notes: definitions, solved examples, code and important questions

**Program:** B.Tech | **Branches:** ALL | **Semester:** 6 | **L-T-P:** 2-0-0

**Scheme:** AKTU B.Tech 3rd year (semester 6), Non-credit (audit) course, 2024-25 onwards (earlier code KNC602, "Indian Tradition, Culture and Society")

**UNIT 1**
Society, State and Polity in India

**UNIT 2**
Indian Literature, Culture, Tradition and Practices

**UNIT 3**
Indian Religion, Philosophy and Practices

**UNIT 4**
Science, Management and Indian Knowledge System

**UNIT 5**
Cultural Heritage and Performing Arts

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: the main ideas, arranged in tables and lists.
3. **Examples and comparisons**: the facts and contrasts that make a long answer stand out.
4. **Key points to remember**: names, dates and terms in one place.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

## How to use these notes the day before the paper

- **Morning (2 hours):** Unit 1 (theories of the state, saptanga, purusharthas, varna and ashrama, position of women) and Unit 2 (scripts, Vedic and epic literature, Buddhist and Jain texts, Arthashastra, regional literatures).
- **Afternoon (2 hours):** Unit 3 (Vedic religion, Buddhism, Jainism, six darshanas, Advaita, Bhakti and Sufi, reform movements) and Unit 4 (mathematics, astronomy, medicine, metallurgy, water management, trade).
- **Evening (1 hour):** Unit 5 (temple architecture, sculpture, painting, dance, music, theatre, festivals, cinema). Then the "Last-hour checklist".
- **In the exam:** name the text, the scholar or the site, give the approximate period, and link the old idea to a modern one where you can (for example the Harappan drainage system and modern urban planning). Write in points with short headings.

**AKTU paper pattern (theory, 70 marks, 3 hours):** Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own

Section C question, so do not skip any unit. This is a non-credit course, but you must pass it to get your degree.

# UNIT 1 Society, State and Polity in India

## ✍️ 2-mark definitions

- **State:** a political community with four elements: population, territory, government and sovereignty.
- **Rajya / Rashtra:** ancient Indian terms for the state and the kingdom.
- **Saptanga theory:** Kautilya's theory that the state has seven limbs (anga): Swami (king), Amatya (ministers), Janapada (territory and people), Durga (fort), Kosha (treasury), Danda (army and justice) and Mitra (ally).
- **Mandala theory:** Kautilya's theory of foreign relations. A king (vijigishu) is surrounded by a circle of states; the immediate neighbour is a natural enemy and the neighbour's neighbour a natural friend.
- **Matsya nyaya:** "the law of the fish", where the big fish eat the small. It describes the chaos of a society without a king, and is used to justify the need for a state.
- **Sabha and Samiti:** popular assemblies of the Vedic period. The Sabha was a smaller council of elders; the Samiti was a general assembly of the people.
- **Mantriparishad:** the council of ministers that advised the king.
- **Purusharthas:** the four aims of human life: Dharma (duty and righteousness), Artha (wealth and material well-being), Kama (desire and pleasure) and Moksha (liberation).
- **Varna:** the fourfold division of society in Vedic texts: Brahmana, Kshatriya, Vaishya and Shudra. Originally linked to occupation and duty.
- **Jati:** a hereditary, endogamous (marrying within the group) occupational group; the basis of the later caste system.
- **Ashrama:** the four stages of life: Brahmacharya (student), Grihastha (householder), Vanaprastha (retiring to the forest) and Sannyasa (renunciation).
- **Samskaras:** life-cycle rites from birth to death; the Grihya Sutras list sixteen of them, including marriage (vivaha).

## 💡 Core concepts

### Ancient Indian theories of the origin of the state

| Theory | Main idea | Source in Indian texts |
|---|---|---|
| Evolutionary (natural growth) | The state grew gradually from the family, clan (kula), tribe (jana) and village into the kingdom (janapada) | Vedic texts show this change from tribe to territorial kingdom |
| Force | The state came from war and conquest; the strongest became king | Aitareya Brahmana: the gods chose Indra as king to lead them in war against the asuras |
| Mystical / divine | The king is divine or created by god to protect people and uphold dharma | Manusmriti says the Creator made the king from parts of eight deities (Indra, Vayu, Yama, Surya, Agni, Varuna, Chandra, Kubera) |

| Theory | Main idea | Source in Indian texts |
|---|---|---|
| Contract (social contract) | People in chaos (matsya nyaya) agreed to accept a king, who received taxes in return for protection | Mahabharata (Shanti Parva): the people chose Manu as king; Buddhist Digha Nikaya: the people elected Mahasammata ("great elect") |

### Stages of state formation in ancient India

1. **Rig Vedic tribal polity (c. 1500 to 1000 BCE):** jana (tribe) led by a rajan, with assemblies (sabha, samiti, vidatha). The king's power was limited; wealth was mostly cattle.
2. **Later Vedic period (c. 1000 to 600 BCE):** the tribe settled on territory (janapada). Kingship became hereditary; rituals such as Rajasuya and Ashvamedha increased the king's power; assemblies declined.
3. **Mahajanapadas (6th century BCE):** sixteen large states (for example Magadha, Kosala, Vatsa, Avanti), some monarchies and some republics (ganas or sanghas, such as the Vajji confederacy and the Shakyas).
4. **Empire (from 4th century BCE):** the Mauryan Empire (Chandragupta, Ashoka) with a centralized bureaucracy, as described in the Arthashastra and by Megasthenes in Indica. Later the Guptas had a more decentralized system with feudal features.

### Kingship, council of ministers and administration

- **Duties of the king (rajadharma):** protect the people (raksha), uphold dharma, punish wrongdoers (danda), collect fair taxes (bhaga, usually one-sixth of produce), promote welfare. Kautilya: "In the happiness of his subjects lies the king's happiness."
- **Council of ministers:** the king should not rule alone ("a single wheel cannot move a chariot"). Kautilya mentions mantrins (inner council) and a larger mantriparishad. Officers included the Purohita (priest), Senapati (commander), Samaharta (chief tax collector) and Sannidhata (treasurer).
- **Administration:** province, district and village units; village headman (gramani); spies (guda purusha); the Mauryan city administration of Pataliputra had six committees of five members each (according to Megasthenes).

### Political ideals and conditions for the welfare of societies

- The king is bound by dharma; he is a servant of the people, paid by taxes.
- Seven limbs (saptanga) must all be strong, with the king most important.
- Buddha's seven conditions for the welfare of the Vajji republic (Mahaparinibbana Sutta): meet often and in large numbers; meet, act and disperse in harmony; follow established laws; respect elders; protect women; honour shrines; protect holy people.
- Ashoka's dhamma: tolerance, non-violence, care for animals, welfare works (roads, wells, rest houses, hospitals), and officers (dhamma mahamattas) to spread these values.

### Society: purusharthas, varnashrama and marriage

- Purusharthas balance worldly life (artha, kama) with ethics (dharma) and the spiritual goal (moksha). Dharma regulates artha and kama.
- Varnashrama dharma combined the varna system with the four ashramas, each with its duties. Over time, varna became rigid and birth-based (jati); this led to discrimination and untouchability, which later reformers and the Constitution (Article 17) rejected.

- **Eight forms of marriage (Manusmriti):** Brahma, Daiva, Arsha and Prajapatya (approved forms); Asura (by payment), Gandharva (by mutual love), Rakshasa (by capture) and Paishacha (by deceit) (disapproved forms). Swayamvara (choice by the bride) is also mentioned in the epics.

### Gender and the position of women

- **Rig Vedic period:** relatively high status. Women took part in assemblies and rituals, received education, and some composed hymns (Ghosha, Lopamudra, Apala, Vishvavara). Philosophers Gargi and Maitreyi debate in the Upanishads.
- **Later period:** status declined: early marriage, restrictions on education and property, sati in some regions, purdah later on. Manusmriti says a woman should be protected by her father, husband and son. However, the same text also says "where women are honoured, the gods rejoice".
- **Representation and challenges:** women rulers and scholars appear (Prabhavati Gupta, Rudramadevi of the Kakatiyas, Ahilyabai Holkar later), and Buddhism and Jainism admitted women as nuns. Challenges included dowry, unequal property rights and lack of education, which 19th-century reformers and modern law addressed.
- **Gender as a social category:** roles of men and women are shaped by society, not only by biology; texts describe ideal roles that differed by region, period and class.

### Four-class classification and slavery

- **Four classes (varna):** Brahmana (teaching, priesthood), Kshatriya (protection, government), Vaishya (agriculture, trade, cattle) and Shudra (service). The Purusha Sukta of the Rig Veda (Book 10) describes them as coming from different parts of the cosmic being.
- **Slavery (dasa):** existed, but in a milder form than in Greece or Rome. Kautilya's Arthashastra lists types of slaves and gives them rights: an Arya could not be made a permanent slave, a slave could own property and buy freedom, and mistreatment was punished. Megasthenes wrongly reported that there were no slaves in India, which shows how mild the practice appeared to outsiders.

## Examples and comparisons

- **Kautilya vs Hobbes:** both describe chaos without a ruler (matsya nyaya vs "the war of all against all") and a ruler accepted to bring order. Kautilya adds that the king remains bound by dharma and must work for the people's welfare.
- **Ancient republics:** the Vajji (Licchavi) gana had an assembly, voting by tokens (shalaka), and a quorum rule, which are often cited as early democratic practices.
- **Saptanga vs modern state:** Swami = executive; Amatya = bureaucracy; Janapada = territory and population; Durga = defence infrastructure; Kosha = finance; Danda = army and police; Mitra = foreign alliances.

## Key points to remember

- Saptanga: Swami, Amatya, Janapada, Durga, Kosha, Danda, Mitra.
- Theories: evolutionary, force (Aitareya Brahmana), divine (Manusmriti), contract (Mahabharata Shanti Parva; Mahasammata in Digha Nikaya).
- Purusharthas: dharma, artha, kama, moksha. Ashramas: brahmacharya, grihastha, vanaprastha, sannyasa.
- Eight forms of marriage; Vedic women scholars Gargi, Maitreyi, Lopamudra, Ghosha, Apala.

## ⭐ Important questions

1. Explain the ancient Indian theories of the origin of the state.
2. Describe Kautilya's saptanga theory and compare it with the elements of a modern state.
3. Trace the stages of state formation from the Vedic tribe to the Mauryan Empire. Explain kingship and the council of ministers.
4. Explain the purusharthas and the varnashrama system.
5. Discuss the position of women in ancient Indian society.
6. Write notes on: forms of marriage; slavery in ancient India; Buddha's conditions for the welfare of a republic.

# UNIT 2 Indian Literature, Culture, Tradition and Practices

## ✍️ 2-mark definitions

- **Harappan script:** the script found on seals of the Indus Valley Civilisation (c. 2600 to 1900 BCE). It has about 400 signs, is usually written right to left, and has not yet been deciphered.
- **Brahmi script:** the ancestor of most Indian and South-East Asian scripts (Devanagari, Bengali, Tamil, Telugu and others). It is written left to right; its earliest clear examples are Ashoka's edicts (3rd century BCE). James Prinsep deciphered it in 1837.
- **Kharosthi:** a script written right to left, used in the north-west (Gandhara); some of Ashoka's edicts use it.
- **Shruti:** "that which is heard": the revealed texts (the Vedas, including Samhitas, Brahmanas, Aranyakas and Upanishads).
- **Smriti:** "that which is remembered": texts of human authorship (Dharmashastras, epics, Puranas).
- **Vedas:** the four oldest sacred texts: Rig Veda (hymns), Sama Veda (melodies for chanting), Yajur Veda (sacrificial formulas) and Atharva Veda (spells, medicine, daily life).
- **Upanishads:** philosophical texts at the end of the Vedas (Vedanta), discussing Brahman (ultimate reality) and Atman (self). About 108 are known; 10 to 13 are considered principal.
- **Itihasa:** "so it was": the two epics, the Ramayana (by Valmiki) and the Mahabharata (attributed to Vyasa).
- **Puranas:** texts of myths, genealogies and religious practice; 18 major Puranas (for example Vishnu, Bhagavata, Shiva, Matsya, Markandeya).
- **Tripitaka:** "three baskets", the Buddhist canon in Pali: Vinaya Pitaka (monastic rules), Sutta Pitaka (Buddha's teachings) and Abhidhamma Pitaka (philosophy).
- **Agamas:** the sacred texts of Jainism, written in Ardhamagadhi Prakrit.
- **Sangam literature:** the earliest Tamil literature (c. 300 BCE to 300 CE), composed at three assemblies (sangams) of poets in Madurai.

## 💡 Core concepts

### Evolution of scripts and languages

- Harappan script (undeciphered) → Brahmi and Kharosthi (Mauryan period) → Gupta script → Siddhamatrika → Nagari and Devanagari in the north; Grantha, Tamil, Telugu-Kannada and Malayalam scripts in the south.
- Languages: Vedic Sanskrit → Classical Sanskrit (Panini's grammar, Ashtadhyayi, c. 5th to 4th century BCE) and Prakrits (spoken languages such as Pali, Ardhamagadhi, Shauraseni) → Apabhramsha → modern Indo-Aryan languages (Hindi, Bengali, Marathi, Gujarati and others). The Dravidian family (Tamil, Telugu, Kannada, Malayalam) developed separately, with Tamil the oldest literary language among them.
- India today recognizes 22 languages in the Eighth Schedule, and 11 have been declared classical languages (including Tamil, Sanskrit, Telugu, Kannada, Malayalam and Odia).

### Vedic literature

| Part | Content |
|---|---|
| Samhitas | Collections of hymns and mantras (the four Vedas) |
| Brahmanas | Prose explaining rituals (for example Aitareya, Shatapatha) |
| Aranyakas | "Forest texts", reflecting on the meaning of rituals |
| Upanishads | Philosophy of Brahman and Atman; "Tat tvam asi" (Chandogya), "Aham Brahmasmi" (Brihadaranyaka) |

- Vedangas (six limbs for understanding the Vedas): Shiksha (phonetics), Kalpa (rituals), Vyakarana (grammar), Nirukta (etymology), Chhandas (metre), Jyotisha (astronomy).
- India's national motto "Satyameva Jayate" comes from the Mundaka Upanishad.

### Epics and Puranas

- **Ramayana:** 24,000 verses in seven kandas. Ideals of dharma, the ideal king (Ram Rajya), duty and family values. Retold in many languages: Kamba Ramayanam (Tamil), Ramcharitmanas by Tulsidas (Awadhi), Krittivasi Ramayana (Bengali).
- **Mahabharata:** about 100,000 verses in 18 parvas, the longest epic in the world. Contains the Bhagavad Gita (in the Bhishma Parva), the dialogue between Krishna and Arjuna on duty, action without attachment (nishkama karma), and the paths of knowledge, action and devotion.
- **Puranas:** five subjects (pancha lakshana): creation, re-creation, genealogy of gods and sages, cosmic ages (manvantaras) and dynasties of kings. They spread religion to the common people through stories.

### Buddhist and Jain literature

- **Buddhist:** Tripitaka in Pali; Jataka tales (stories of the Buddha's previous births); Milindapanha (dialogue between King Menander and the monk Nagasena); Dipavamsa and Mahavamsa (Sri Lankan chronicles). In Sanskrit: Ashvaghosha's Buddhacharita, Nagarjuna's works on Madhyamaka philosophy.
- **Jain:** 12 Angas and other Agamas in Prakrit; Kalpa Sutra (by Bhadrabahu); later Sanskrit works like Hemachandra's Parishishtaparvan.

### Kautilya's Arthashastra and major Sanskrit authors

- **Arthashastra** (attributed to Kautilya, also called Chanakya or Vishnugupta, minister of Chandragupta Maurya): 15 books on statecraft, economics, law, administration, diplomacy, espionage, war and taxation. Rediscovered by R. Shamasastry in 1905.
- **Sanskrit authors:** Panini (grammar), Patanjali (Mahabhashya and Yoga Sutras), Bhasa (plays such as Svapnavasavadattam), Kalidasa (Abhijnanashakuntalam, Meghadutam, Raghuvamsham, Kumarasambhavam), Shudraka (Mricchakatika), Vishakhadatta (Mudrarakshasa), Banabhatta (Harshacharita, Kadambari), Bharavi (Kiratarjuniyam), Magha (Shishupalavadha), Jayadeva (Gita Govinda), Kalhana (Rajatarangini, history of Kashmir), Vishnu Sharma (Panchatantra).

### Regional literatures of the south

- **Sangam (Tamil):** Ettuthogai (eight anthologies) and Pattuppattu (ten idylls); themes of akam (love, inner life) and puram (war, public life). Tolkappiyam is the oldest Tamil grammar. Later works:

Tirukkural by Thiruvalluvar (ethics, in three parts on dharma, wealth and love); the epics Silappatikaram (Ilango Adigal) and Manimekalai.

- **Telugu:** Nannaya began the Telugu Mahabharata (11th century), completed by Tikkana and Errana. Golden age under Krishnadevaraya of Vijayanagara (16th century), whose court had the Ashtadiggajas (eight poets) including Allasani Peddana; Krishnadevaraya wrote Amuktamalyada. Tenali Ramakrishna, Vemana (moral verses).
- **Kannada:** Kavirajamarga (9th century, by Amoghavarsha's court) is the earliest known work; the three gems (ratnatraya) Pampa, Ponna and Ranna; the Vachana literature of Basavanna and Akka Mahadevi (12th century); Kumaravyasa's Mahabharata.
- **Malayalam:** Ramacharitam (early work), Thunchaththu Ezhuthachan (16th century, called the father of Malayalam literature; Adhyatma Ramayanam), Kunchan Nambiar (Ottan Thullal).

**Northern languages and literature**

- **Persian:** became the court language under the Delhi Sultanate and the Mughals. Amir Khusrau (13th to 14th century, poet, musician, called the "parrot of India"); Babur's Baburnama (actually in Turkish, translated into Persian); Abul Fazl's Akbarnama and Ain-i-Akbari; Persian translations of the Mahabharata (Razmnama) and Upanishads (Sirr-i-Akbar by Dara Shikoh).
- **Urdu:** developed from Hindavi with Persian, Arabic and Turkish words in the Deccan and Delhi. Poets: Wali Dakhni, Mir Taqi Mir, Mirza Ghalib, Allama Iqbal; prose: Premchand also wrote in Urdu.
- **Hindi:** early forms in the Siddha and Nath poets and Prithviraj Raso (Chand Bardai). Bhakti period: Kabir (dohas), Surdas (Sursagar, Braj), Tulsidas (Ramcharitmanas, Awadhi), Mirabai, Malik Muhammad Jayasi (Padmavat). Modern: Bharatendu Harishchandra (father of modern Hindi), Premchand, Jaishankar Prasad, Suryakant Tripathi "Nirala", Mahadevi Varma, Sumitranandan Pant (Chhayavad poets).

## Examples and comparisons

- **Shruti vs Smriti:** Shruti is considered revealed and eternal (Vedas); Smriti is remembered and human-authored, and can change with time (Manusmriti, epics, Puranas). Where they conflict, Shruti prevails.
- **One story, many languages:** the Ramayana has hundreds of versions, including Jain and Buddhist versions and versions in Thailand (Ramakien) and Indonesia (Kakawin Ramayana), which shows India's cultural influence.
- **Tirukkural** has been translated into many languages and is often called a universal book of ethics.

## Key points to remember

- Harappan script undeciphered; Brahmi deciphered by James Prinsep (1837).
- Four Vedas; six Vedangas; Upanishads = Vedanta; "Satyameva Jayate" from Mundaka Upanishad.
- Tripitaka (Pali): Vinaya, Sutta, Abhidhamma. Jain Agamas (Prakrit).
- Arthashastra: Kautilya; rediscovered by Shamasastry (1905).
- Kalidasa's works; Sangam: Tolkappiyam, Tirukkural, Silappatikaram, Manimekalai.
- Amir Khusrau, Ghalib, Kabir, Tulsidas, Surdas, Premchand.

## ⭐ Important questions

1. Trace the evolution of scripts and languages in India from the Harappan script to modern scripts.
2. Describe Vedic literature: the Vedas, Brahmanas, Aranyakas, Upanishads and Vedangas.
3. Explain the importance of the Ramayana, the Mahabharata and the Puranas in Indian culture.
4. Write a note on Buddhist and Jain literature. Discuss Kautilya's Arthashastra.
5. Describe the major Sanskrit authors and their works.
6. Write notes on Sangam literature and on the Telugu, Kannada and Malayalam literatures.
7. Discuss the growth of Persian, Urdu and Hindi literature.

# UNIT 3 Indian Religion, Philosophy and Practices

## ✍️ 2-mark definitions

- **Pre-Vedic religion:** the religious practices of the Harappan people, known only from archaeology: mother goddess figurines, the "Pashupati" seal (a horned figure in a yogic posture surrounded by animals), worship of trees (pipal), animals and possibly fire (fire altars at Kalibangan and Lothal), ritual bathing (Great Bath of Mohenjo-daro).
- **Rita:** the Vedic idea of cosmic order, which the god Varuna upholds; the forerunner of dharma.
- **Yajna:** Vedic fire sacrifice with offerings to gods through Agni.
- **Karma:** the principle that every action has consequences that shape future experience and rebirth.
- **Samsara:** the cycle of birth, death and rebirth.
- **Moksha / Nirvana:** liberation from samsara. Nirvana in Buddhism means the extinction of craving and suffering.
- **Darshana:** "a way of seeing": a school of Indian philosophy.
- **Astika and Nastika:** astika (orthodox) schools accept the authority of the Vedas; nastika (heterodox) schools reject it (Buddhism, Jainism, Charvaka, Ajivika).
- **Advaita Vedanta:** the non-dualist philosophy of Adi Shankaracharya: Brahman alone is real, the world is maya (appearance), and the individual self (Atman) is identical with Brahman.
- **Bhakti:** loving devotion to a personal god as the path to liberation, open to all regardless of caste or gender.
- **Sufism:** the mystical tradition in Islam that seeks direct experience of God through love, music (sama) and the guidance of a teacher (pir).
- **Silsila:** a Sufi order (lineage), such as Chishti, Suhrawardi, Qadiri and Naqshbandi.

## 💡 Core concepts

### Vedic religion

- Early Vedic gods were nature powers: Indra (rain, war; most hymns), Agni (fire, messenger between humans and gods), Varuna (cosmic order), Soma (sacred drink), Surya, Ushas (dawn), Vayu.
- Worship was by hymns and sacrifices (yajna) for material gains such as cattle, sons and victory. No temples or idols.
- In the later Vedic period rituals became elaborate and priests powerful; Prajapati, Vishnu and Rudra rose in importance. The Upanishads reacted against ritualism and turned to knowledge (jnana), introducing karma, samsara, moksha, Brahman and Atman.

### Buddhism (Gautama Buddha, c. 563 to 483 BCE, by the traditional dating)

- Born at Lumbini; enlightenment at Bodh Gaya; first sermon at Sarnath ("turning the wheel of dharma"); death (mahaparinirvana) at Kushinagar.
- **Four Noble Truths:** life involves suffering (dukkha); suffering has a cause (craving, tanha); suffering can end (nirodha); there is a path to end it (magga).

- **Eightfold Path:** right view, right intention, right speech, right action, right livelihood, right effort, right mindfulness, right concentration. Grouped as wisdom (prajna), ethics (shila) and meditation (samadhi). It is the "Middle Path" between luxury and extreme austerity.
- Other teachings: impermanence (anicca), no permanent self (anatta), dependent origination (pratitya samutpada), karma without a creator god.
- **Councils:** First (Rajagriha), Second (Vaishali), Third (Pataliputra, under Ashoka), Fourth (Kashmir, under Kanishka; split into Mahayana and Hinayana).
- **Schools:** Theravada/Hinayana (Buddha as teacher, individual liberation, Pali), Mahayana (Buddha as divine, bodhisattva ideal of helping all beings, Sanskrit), Vajrayana (tantric, Tibet).
- Spread by Ashoka's missionaries to Sri Lanka and South-East Asia; Mahayana went along the Silk Road to China, Korea and Japan.

### Jainism (Vardhamana Mahavira, 24th Tirthankara, c. 6th century BCE)

- Rishabhadeva was the first Tirthankara and Parshvanatha the 23rd. Mahavira attained kevala jnana (omniscience) and is called Jina ("conqueror"), hence "Jain".
- **Three jewels (triratna):** right faith, right knowledge, right conduct.
- **Five vows (mahavratas):** ahimsa (non-violence), satya (truth), asteya (non-stealing), aparigraha (non-possession), brahmacharya (chastity, added by Mahavira).
- Philosophy: every living being has a soul (jiva); anekantavada (reality has many aspects), syadvada (every statement is true only from some point of view).
- Sects: Digambara ("sky-clad", monks wear nothing) and Shvetambara ("white-clad").

### Six orthodox schools (shad darshana)

| School | Founder / text | Main idea |
|---|---|---|
| Nyaya | Gautama (Akshapada), Nyaya Sutra | Logic and valid knowledge; four pramanas: perception, inference, comparison, testimony |
| Vaisheshika | Kanada, Vaisheshika Sutra | Atomism: everything is made of eternal atoms (paramanu); six categories (padarthas) |
| Samkhya | Kapila | Dualism of purusha (consciousness) and prakriti (matter); three gunas: sattva, rajas, tamas |
| Yoga | Patanjali, Yoga Sutras | Eight limbs (ashtanga): yama, niyama, asana, pranayama, pratyahara, dharana, dhyana, samadhi; "yoga is the stilling of the mind's fluctuations" |
| Mimamsa (Purva Mimamsa) | Jaimini | Interpretation of Vedic rituals; duty (dharma) through correct ritual action |
| Vedanta (Uttara Mimamsa) | Badarayana, Brahma Sutra | Knowledge of Brahman, based on the Upanishads, Brahma Sutra and Gita (prasthana traya) |

They form pairs: Nyaya–Vaisheshika, Samkhya–Yoga, Mimamsa–Vedanta.

### Schools of Vedanta and Shankaracharya

- **Adi Shankaracharya (c. 8th century CE, born at Kalady, Kerala):** Advaita (non-dualism). "Brahma satyam jagat mithya, jivo brahmaiva naparah": Brahman is real, the world is an appearance, the self is none other than Brahman. Ignorance (avidya) hides this; knowledge (jnana) liberates. He wrote commentaries on the prasthana traya, debated other schools across India and founded four mathas: Sringeri (south), Dwarka (west), Puri (east) and Jyotirmath (Badrinath, north).

- **Ramanuja (11th to 12th century):** Vishishtadvaita (qualified non-dualism): souls and the world are real, but are the body of God (Vishnu); bhakti leads to liberation.
- **Madhvacharya (13th century):** Dvaita (dualism): God, souls and matter are eternally distinct.
- **Nimbarka:** Dvaitadvaita (dualism in non-dualism). **Vallabhacharya:** Shuddhadvaita (pure non-dualism), Pushtimarga.

### Heterodox sects

- **Charvaka (Lokayata):** materialism; only perception is valid knowledge; denies soul, afterlife, god and the Vedas; "eat, drink and be merry".
- **Ajivika (Makkhali Gosala):** strict determinism (niyati); everything is predetermined by fate.
- Buddhism and Jainism (above).

### Bhakti movement

- Began in South India with the **Alvars** (Vishnu devotees; their hymns are collected in the Nalayira Divya Prabandham) and the **Nayanars** (Shiva devotees; Tevaram), between the 6th and 9th centuries CE. Spread north from about the 13th century.
- **Features:** devotion to one god, rejection of empty ritual and caste discrimination, use of local languages, equal place for women, the guru's guidance.
- **Two streams:** Saguna (god with form: Rama and Krishna devotees such as Tulsidas, Surdas, Mirabai, Chaitanya Mahaprabhu) and Nirguna (formless god: Kabir, Guru Nanak, Ravidas, Dadu).
- **Regional saints:** Maharashtra: Jnaneshwar, Namdev, Eknath, Tukaram; Karnataka: Basavanna (Virashaivism, Lingayats); Bengal: Chaitanya; Assam: Shankardev; Punjab: Guru Nanak (founder of Sikhism; teachings in the Guru Granth Sahib).

### Sufi movement

- Came with Muslim mystics from Persia and Central Asia from about the 11th century. Main orders in India: **Chishti** (Khwaja Moinuddin Chishti at Ajmer, Nizamuddin Auliya and Bakhtiyar Kaki in Delhi, Salim Chishti at Fatehpur Sikri), **Suhrawardi** (Multan), **Qadiri** and **Naqshbandi** (more orthodox).
- Teachings: love of God, service to humanity, tolerance, simple living, music (qawwali), the khanqah (hospice) open to all.
- Bhakti and Sufi movements together promoted harmony between Hindus and Muslims and enriched regional languages and music.

### 19th-century socio-religious reform movements

| Movement | Founder (year) | Main work |
|---|---|---|
| Brahmo Samaj | Raja Ram Mohan Roy (1828) | Monotheism, against idol worship; campaign that led to the abolition of sati (1829); modern education |
| Prarthana Samaj | Atmaram Pandurang (1867), with M. G. Ranade | Widow remarriage, women's education, against caste |
| Arya Samaj | Swami Dayananda Saraswati (1875) | "Back to the Vedas", Satyarth Prakash, against idol worship and caste by birth, DAV schools, shuddhi |

| Movement | Founder (year) | Main work |
|---|---|---|
| Ramakrishna Mission | Swami Vivekananda (1897) | Universal religion, service to humanity as worship, Chicago address (1893) |
| Theosophical Society | Blavatsky and Olcott (1875); in India Annie Besant | Revival of Hindu and Buddhist thought |
| Satyashodhak Samaj | Jyotiba Phule (1873) | Rights of lower castes and women; Savitribai Phule opened girls' schools |
| Aligarh Movement | Sir Syed Ahmed Khan (1875, MAO College) | Modern education for Muslims |
| Self-Respect Movement | E. V. Ramasamy "Periyar" (1925) | Against caste and rituals in Tamil Nadu |

Other reformers: Ishwar Chandra Vidyasagar (Widow Remarriage Act, 1856), Narayana Guru in Kerala ("one caste, one religion, one god for humankind"), Pandita Ramabai.

**Modern religious practices:** pilgrimage (tirtha yatra, Kumbh Mela, Char Dham, Hajj, Amritsar), yoga and meditation spreading worldwide (International Day of Yoga, 21 June), festivals as community events, online worship, interfaith dialogue, and the constitutional principle of secularism (equal respect for all religions, Articles 25 to 28).

## Examples and comparisons

- **Buddhism vs Jainism:** both reject Vedic authority and caste, preach non-violence and karma. Buddhism follows the Middle Path and anatta (no permanent self); Jainism stresses extreme austerity and accepts a soul in every being. Buddhism spread widely abroad; Jainism stayed mostly in India.
- **Advaita vs Dvaita:** Advaita says the self and Brahman are one; Dvaita says they are always separate, and devotion to God is the way.
- **Bhakti vs Sufi:** both emphasise love of God, equality and the teacher; Bhakti grew within Hindu traditions and Sufism within Islam; both used music and local languages.

## Key points to remember

- Pashupati seal and Great Bath (Harappan religion).
- Four Noble Truths and Eightfold Path; four Buddhist councils.
- Triratna and five mahavratas of Jainism; Digambara and Shvetambara.
- Six darshanas with founders; three pairs.
- Shankara's four mathas; Ramanuja (Vishishtadvaita); Madhva (Dvaita).
- Alvars and Nayanars; Kabir, Nanak, Tulsidas, Mirabai, Chaitanya; Chishti saints.
- Reform movements table.

## ⭐ Important questions

1. Describe the religion of the Harappan and Vedic periods.
2. Explain the teachings of Buddhism and Jainism and compare them.

3. Explain the six orthodox schools of Indian philosophy.
4. Discuss the philosophy of Adi Shankaracharya and compare it with other schools of Vedanta.
5. Write notes on heterodox sects: Charvaka and Ajivika.
6. Explain the Bhakti and Sufi movements and their impact on Indian society.
7. Describe the socio-religious reform movements of the 19th century.

# UNIT 4 Science, Management and Indian Knowledge System

## ✍️ 2-mark definitions

- **Indian Knowledge System (IKS):** the body of knowledge developed in India over thousands of years in science, mathematics, medicine, arts, governance and philosophy, passed on through texts and traditions.
- **Shulba Sutras:** Vedic texts (c. 800 to 200 BCE) giving geometry for building fire altars, including a statement of the result now called the Pythagoras theorem and an approximation of $\sqrt{2}$.
- **Ayurveda:** the traditional Indian system of medicine, based on the balance of three doshas (vata, pitta, kapha). Its main texts are the Charaka Samhita (internal medicine) and the Sushruta Samhita (surgery).
- **Rasashastra:** the Indian tradition of alchemy and chemistry, especially preparation of medicines from metals and minerals (Nagarjuna's Rasaratnakara).
- **Wootz steel:** high-carbon crucible steel made in South India from about 300 BCE, exported and famous as the material of "Damascus" swords.
- **Stepwell (baoli / vav):** a well with steps down to the water, used for storage, cooling and community gathering; for example Rani ki Vav, Patan (UNESCO World Heritage Site).
- **Dholavira:** a Harappan city in Gujarat with an advanced water-conservation system of reservoirs and channels (UNESCO site, 2021).
- **Kalpa:** Vedanga dealing with rituals; it includes the Shulba Sutras.

## 💡 Core concepts

### Mathematics

- **Zero and the decimal place-value system:** the place-value system with zero was developed in India. The Bakhshali manuscript uses a dot for zero. Brahmagupta (Brahmasphutasiddhanta, 628 CE) gave rules for arithmetic with zero and negative numbers. These numerals reached Europe through the Arabs ("Hindu-Arabic numerals").
- **Aryabhata (476 to c. 550 CE), Aryabhatiya:** value of $\pi \approx 3.1416$ (stated as 62832/20000), sine tables, solutions of indeterminate equations (kuttaka), sum of series.
- **Brahmagupta:** solution of quadratic equations, Brahmagupta's formula for the area of a cyclic quadrilateral, rules for zero.
- **Bhaskara II (1114 to 1185), Lilavati and Bijaganita:** arithmetic, algebra, solutions of Pell's equation (chakravala method), early ideas of differential calculus.
- **Kerala school (Madhava of Sangamagrama, 14th century; Nilakantha, Jyesthadeva):** infinite series for $\pi$, sine and cosine, about 250 years before Newton and Leibniz.
- **Pingala (Chhandas Sutra):** binary-like representation of metres and a pattern related to Pascal's triangle (meru prastara). **Hemachandra** described the sequence later called the Fibonacci sequence.

### Astronomy

- Vedanga Jyotisha (earliest astronomy text, for calendar and ritual timing).

- Aryabhata: the Earth rotates on its axis; explained solar and lunar eclipses by shadows, not by Rahu and Ketu; length of the year close to the modern value.
- Varahamihira (6th century), Pancha Siddhantika (summary of five astronomical systems) and Brihat Samhita (an encyclopaedia including weather, architecture and gems).
- Observatories (Jantar Mantar) built by Sawai Jai Singh II in the 18th century at Jaipur, Delhi, Ujjain, Varanasi and Mathura; the Jaipur one is a UNESCO site.
- India's satellites Aryabhata (1975) and Bhaskara were named after these astronomers.

### Physics and chemistry

- **Kanada (Vaisheshika):** everything is made of indivisible atoms (paramanu) that combine into dyads and triads; ideas of motion and gravity (gurutva).
- **Chemistry:** Harappan faience and glazed pottery; natural dyes (indigo, madder); perfumes and distillation; Nagarjuna's rasashastra (purifying mercury, preparing metallic medicines called bhasmas); cosmetics and medicines; the Delhi Iron Pillar's protective coating.

### Medicine and biology

- **Charaka Samhita:** diagnosis, internal medicine, physiology, diet, preventive health and an oath for physicians.
- **Sushruta Samhita (Sushruta, "father of surgery"):** describes more than 300 operations and 120 surgical instruments, cataract surgery and plastic surgery (rhinoplasty, rebuilding the nose with a forehead flap), and the training of surgeons on models.
- Other systems: Siddha (Tamil Nadu), Unani (Greek-Arab, developed in India), Yoga for health. The Ministry of AYUSH promotes these today.
- **Biology:** classification of plants and animals in Ayurvedic texts; Vrikshayurveda (science of plants) by Surapala on plant health, grafting and irrigation; veterinary texts on horses (Ashvashastra) and elephants (Hastyayurveda).

**Geography:** the Puranas describe the world as continents (dvipas) with Jambudvipa (including Bharatavarsha) at the centre. Ancient Indians had detailed knowledge of rivers, mountains and trade routes; Varahamihira discussed monsoon prediction.

### Agriculture and technology

- **Agriculture:** Harappan ploughed field at Kalibangan (criss-cross furrows); crops: wheat, barley, rice, cotton (Harappans were the first known to grow cotton), sugarcane. Krishi Parashara (text on agriculture and weather), Kautilya's Arthashastra on irrigation and seeds; crop rotation, mixed cropping, organic manure.
- **Metallurgy:** copper and bronze (Harappan "Dancing Girl" made by the lost-wax method); **Delhi Iron Pillar** (Gupta period, about 1,600 years old, 98% pure wrought iron, rust-resistant because of a protective film of iron hydrogen phosphate formed by its high phosphorus content); wootz steel; zinc distillation at Zawar, Rajasthan (from about the 12th century; India was the first to smelt zinc by distillation); the Sultanganj Buddha in copper.
- **Harappan technologies:** planned cities on a grid with streets at right angles; standardized burnt bricks in the ratio 1 : 2 : 4; covered drainage connected to every house; granaries; dockyard at Lothal; standard weights (binary and decimal) and measuring scales; seals; bead making.
- **Water management:** Harappan reservoirs (Dholavira) and the Great Bath; Sudarshana lake (built under Chandragupta Maurya, repaired under Rudradaman and Skandagupta, recorded in the

Junagadh inscription); Grand Anicut (Kallanai) on the Kaveri built by the Chola king Karikala (about 2nd century CE, still in use); tanks and canals in South India; stepwells in Gujarat and Rajasthan; kuls in the Himalayas; eri tanks in Tamil Nadu; johads in Rajasthan; zabo system in Nagaland.

- **Textile technology:** Harappan cotton cloth; fine Dhaka muslin; dyeing with mordants and indigo; resist dyeing (bandhani, ikat); block printing (calico from Calicut); silk (Kanchipuram, Banaras), Kashmir shawls. Indian textiles dominated world trade until the 18th century.
- **Writing technology:** palm-leaf and birch-bark (bhojpatra) manuscripts; inks from soot and plant gums; copper plates for land grants; stone inscriptions (Ashokan edicts); paper later came from Central Asia; manuscript preservation with neem and turmeric.
- **Pyrotechnics:** Kautilya mentions agniyoga (fire mixtures) for war. Saltpetre-based fireworks and gunpowder were known by the 15th century (fireworks at the Vijayanagara court). Tipu Sultan's iron-cased Mysore rockets (18th century) inspired the British Congreve rockets.

**Trade and economic dominance**

- **Harappan trade** with Mesopotamia (Meluhha in Mesopotamian texts); dockyard at Lothal; seals found in Mesopotamia.
- **Ancient trade routes:** Uttarapatha (north, from Taxila to Pataliputra and Tamralipti) and Dakshinapatha (south); the Silk Road; Indo-Roman trade in spices, pepper, textiles, gems and ivory (Arikamedu, Muziris; Roman coins found in South India). Guilds (shrenis) of merchants and artisans worked like banks and regulated trade.
- **Maritime trade** with South-East Asia (Chola naval expeditions, 11th century), spreading Indian culture (Angkor Wat, Borobudur).
- **Economic dominance:** estimates by economic historian Angus Maddison put India's share of world GDP at roughly a quarter to a third in the first millennium CE, and around a quarter in 1700, the largest economy in the world at times. The share fell to about 4% by 1950 because of colonial policies, de-industrialisation (collapse of handicrafts and textiles against British machine goods) and the drain of wealth described by Dadabhai Naoroji.

**Management in the Indian Knowledge System:** Arthashastra on administration, planning, finance, audit and HR (selection and testing of officials); Bhagavad Gita on duty without attachment (nishkama karma) and leadership; Thirukkural on ethics in governance; guild management and temple administration as early institutions.

## Examples and comparisons

- **Indian contribution vs later Western discovery:** zero and place-value (Brahmagupta, 7th century) vs adoption in Europe (Fibonacci, 1202); Madhava's series for π (14th century) vs Gregory and Leibniz (17th century); plastic surgery (Sushruta) vs modern rhinoplasty (the "Indian method" reported in 1794 in Britain).
- **Delhi Iron Pillar** is a classic engineering example of corrosion resistance; it is studied by metallurgists, including at IIT Kanpur.
- **Harappan town planning vs modern smart city:** grid layout, zoning (citadel and lower town), drainage and standardized bricks are still basic principles of urban planning.

## Key points to remember

- Aryabhata: π ≈ 3.1416, Earth's rotation, eclipses. Brahmagupta: zero rules (628 CE). Bhaskara II: Lilavati. Madhava: infinite series.
- Charaka (medicine), Sushruta (surgery, rhinoplasty). Kanada: atoms.
- Delhi Iron Pillar; wootz steel; zinc at Zawar.
- Harappan bricks 1 : 2 : 4; Lothal dockyard; Dholavira reservoirs; Grand Anicut by Karikala.
- Uttarapatha, Dakshinapatha, Muziris, Arikamedu; shrenis (guilds).

## ⭐ Important questions

1. Discuss India's contributions to mathematics and astronomy.
2. Describe ancient Indian medicine with reference to Charaka and Sushruta.
3. Explain Indian achievements in metallurgy with examples.
4. Describe the technology and town planning of the Harappan civilisation.
5. Write a note on traditional water management systems in India.
6. Discuss textile technology and trade in ancient India and India's economic position before colonial rule.
7. How is the Indian Knowledge System relevant to modern science and management?

NEXT ENG _sheerat_

# UNIT 5 Cultural Heritage and Performing Arts

## ✍️ 2-mark definitions

- **Cultural heritage:** the monuments, art, traditions, languages and knowledge inherited from the past and passed on to future generations. Tangible heritage is physical (temples, sculptures); intangible heritage is living practice (music, dance, festivals).
- **Stupa:** a hemispherical Buddhist mound containing relics, for example the Great Stupa at Sanchi.
- **Chaitya and vihara:** a chaitya is a Buddhist prayer hall; a vihara is a monastery (for example Karle and Ajanta caves).
- **Nagara style:** the North Indian temple style with a curved tower (shikhara) over the sanctum, for example Khajuraho and the Sun Temple at Konark.
- **Dravida style:** the South Indian temple style with a pyramid-shaped tower (vimana) and large gateways (gopurams), for example the Brihadeeswara Temple, Thanjavur.
- **Vesara style:** a mix of Nagara and Dravida, found in the Deccan (Chalukya and Hoysala temples, such as Belur and Halebidu).
- **Vastu Shastra and Shilpa Shastra:** traditional texts on architecture, town planning and sculpture.
- **Natya Shastra:** Bharata Muni's treatise on drama, dance and music (c. 200 BCE to 200 CE). It describes the theory of rasa (aesthetic emotion) and is called the "fifth Veda".
- **Rasa:** the emotional flavour experienced by the audience; Bharata lists eight (shringara, hasya, karuna, raudra, vira, bhayanaka, bibhatsa, adbhuta), and a ninth, shanta (peace), was added later.
- **Raga:** a melodic framework in Indian classical music, associated with a mood and often a time of day.
- **Tala:** the rhythmic cycle in Indian music, for example teental (16 beats).
- **Gharana:** a school or lineage of Hindustani music or Kathak dance with its own style.

## 💡 Core concepts

### Indian architecture and engineering

- **Harappan:** planned cities, the Great Bath, granaries, dockyard.
- **Mauryan:** polished sandstone Ashokan pillars (the Sarnath lion capital is India's national emblem), stupas, the Barabar rock-cut caves.
- **Buddhist rock-cut architecture:** Ajanta (2nd century BCE to 6th century CE, famous for paintings), Ellora (Buddhist, Hindu and Jain caves; the Kailasa Temple, Cave 16, was carved from a single rock from the top down), Karle, Elephanta.
- **Temple architecture (Gupta period onwards):** early Gupta temples (Dashavatara Temple, Deogarh). Nagara: Khajuraho (Chandela), Konark Sun Temple (shaped like a chariot with 24 wheels that work as sundials), Lingaraja (Bhubaneswar). Dravida: Shore Temple and rathas at Mahabalipuram (Pallava), Brihadeeswara at Thanjavur (Chola, Rajaraja I, 1010 CE; the vimana is about 66 m high, built of granite with interlocking joints and no binding mortar), Meenakshi Temple, Madurai. Vesara: Hoysala temples at Belur and Halebidu.

- **Indo-Islamic architecture:** arches, domes, minarets, calligraphy and gardens: Qutub Minar, Alai Darwaza, Gol Gumbaz (Bijapur, whispering gallery), Humayun's Tomb, Fatehpur Sikri, Taj Mahal (charbagh garden, double dome, white marble with pietra dura inlay), Red Fort.
- **Engineering principles:** load-bearing stone construction, corbelling, interlocking joints, use of iron clamps, earthquake-resistant plinths, natural ventilation and cooling (jaalis, courtyards, stepwells), acoustics (Gol Gumbaz), solar orientation (Konark, Modhera Sun Temple).

### Sculpture, seals, coins, pottery and painting

- **Sculpture:** Harappan "Dancing Girl" (bronze) and "Priest King" (steatite); Mauryan Didarganj Yakshi; Gandhara school (Greek-influenced, grey schist, realistic Buddha images), Mathura school (red sandstone, indigenous), Amaravati school (white marble, narrative reliefs); Gupta Sarnath Buddha (classical perfection); Chola bronzes (Nataraja, made by the lost-wax method).
- **Seals:** Harappan steatite seals with animals (unicorn, bull, elephant) and script, used in trade.
- **Coins:** punch-marked coins (silver, earliest Indian coins, from about the 6th century BCE); Indo-Greek coins with portraits of kings; Kushana gold coins; Gupta gold coins (dinara) showing kings playing the veena or performing the horse sacrifice; Sher Shah Suri's silver rupiya, the origin of the rupee.
- **Pottery:** Harappan black-on-red ware; Ochre Coloured Pottery; Painted Grey Ware (later Vedic); Northern Black Polished Ware (Mauryan, high-quality glossy pottery); terracotta figurines; blue pottery of Jaipur today.
- **Painting:** prehistoric rock paintings at Bhimbetka (UNESCO site); Ajanta murals (Jataka stories, Padmapani); Bagh caves; miniature painting: Pala, Mughal (Akbar's and Jahangir's ateliers), Rajput (Mewar, Kishangarh with "Bani Thani"), Pahari (Basohli, Kangra); folk painting: Madhubani (Bihar), Warli (Maharashtra), Pattachitra (Odisha), Kalamkari (Andhra), Gond (Madhya Pradesh), Tanjore painting.

### Performing arts

- **Classical dance (recognized by the Sangeet Natak Akademi):**

| Dance | State | Feature |
|---|---|---|
| Bharatanatyam | Tamil Nadu | Temple dance; fixed geometric postures; based on Natya Shastra |
| Kathak | Uttar Pradesh (North India) | Storytelling, fast footwork (tatkar), spins (chakkars); Lucknow, Jaipur, Banaras gharanas |
| Kathakali | Kerala | Dance-drama; elaborate face make-up (green for heroes); themes from the epics |
| Kuchipudi | Andhra Pradesh | Dance-drama; dancing on the rim of a brass plate (tarangam) |
| Odissi | Odisha | Tribhanga (three-bend) posture; temple sculpture poses |
| Manipuri | Manipur | Gentle movements; Raas Lila of Krishna |
| Mohiniyattam | Kerala | Graceful, swaying dance by women |
| Sattriya | Assam | Started by Shankardev in Vaishnava monasteries (sattras) |

- **Music:** origins in the Sama Veda. Two traditions: **Hindustani** (North; forms: dhrupad, khayal, thumri, tappa; influenced by Persian music; Tansen at Akbar's court; Amir Khusrau credited with developing the khayal and the sitar and tabla in tradition) and **Carnatic** (South; kriti form; Purandara Dasa called its father; Trinity: Tyagaraja, Muthuswami Dikshitar, Syama Sastri). Instruments: veena, sitar, sarod, flute (bansuri), shehnai, tabla, mridangam. Folk music: baul (Bengal), bhatiali, lavani (Maharashtra), bihu songs (Assam).

- **Theatre and drama:** Sanskrit drama based on the Natya Shastra (Bhasa, Kalidasa, Shudraka); folk theatre: Yakshagana (Karnataka), Jatra (Bengal), Nautanki (Uttar Pradesh), Tamasha (Maharashtra), Bhavai (Gujarat), Ramlila (UNESCO intangible heritage), Koodiyattam (Kerala, the oldest living Sanskrit theatre, UNESCO). Modern: Indian People's Theatre Association (IPTA), National School of Drama.
- **Puppetry:** string puppets (Kathputli of Rajasthan), shadow puppets (Tholu Bommalata of Andhra Pradesh, Togalu Gombeyata of Karnataka, Tolpavakoothu of Kerala), rod puppets (Putul Nach of Bengal), glove puppets (Pavakoothu of Kerala).
- **Martial arts:** Kalaripayattu (Kerala, among the oldest martial arts in the world), Gatka (Punjab), Silambam (Tamil Nadu, stick fighting), Thang-Ta (Manipur), Mallakhamb (Madhya Pradesh and Maharashtra, gymnastics on a pole), Pari-khanda (Bihar).

**Fairs and festivals:** Kumbh Mela (UNESCO intangible heritage, held at Prayagraj, Haridwar, Ujjain and Nashik); Diwali, Holi, Durga Puja (UNESCO intangible heritage, 2021), Eid, Christmas, Gurpurab, Buddha Purnima, Mahavir Jayanti; harvest festivals: Pongal, Makar Sankranti, Baisakhi, Onam, Bihu; fairs: Pushkar camel fair, Sonepur cattle fair, Hornbill festival (Nagaland). Festivals connect agriculture, seasons, religion and community.

**Current developments in arts and culture:** UNESCO World Heritage Sites (India has more than 40, including Dholavira, Ramappa Temple, Santiniketan and the Hoysala temples); intangible heritage listings (yoga, Kumbh Mela, Garba of Gujarat in 2023); the Ministry of Culture, Sangeet Natak Akademi, Lalit Kala Akademi, Sahitya Akademi; digitisation of manuscripts (National Mission for Manuscripts); GI tags for crafts; growth of fusion music, digital art, and streaming platforms for regional content.

**India's cultural contribution to the world:** yoga and meditation; Buddhism in Asia; the decimal system and zero; Ayurveda; the Panchatantra (translated into Persian and Arabic as Kalila wa Dimna and into many European languages); chess (chaturanga); textiles and spices; Indian temple architecture in South-East Asia (Angkor Wat in Cambodia, Borobudur in Indonesia, Prambanan); ideas of non-violence (ahimsa) influencing Gandhi, Martin Luther King Jr. and Nelson Mandela; Indian music (Ravi Shankar's influence on the Beatles); Bollywood and Indian food worldwide.

### Indian cinema

- **Beginnings:** the Lumière brothers' films shown in Bombay (1896); Dadasaheb Phalke's "Raja Harishchandra" (1913), the first full-length Indian feature film (Phalke is called the father of Indian cinema; the Dadasaheb Phalke Award is India's highest film honour); "Alam Ara" (1931), the first Indian sound film.
- **Golden era (1950s):** Satyajit Ray ("Pather Panchali", 1955, part of the Apu trilogy; honorary Oscar 1992), Bimal Roy, Guru Dutt, Raj Kapoor, Mehboob Khan ("Mother India", 1957).
- **Parallel cinema (1970s):** Shyam Benegal, Mrinal Sen, Adoor Gopalakrishnan.
- **Today:** India produces the most films in the world, in many languages (Hindi, Tamil, Telugu, Malayalam, Bengali, Kannada, Marathi). International recognition: "RRR" song "Naatu Naatu" (Oscar for Best Original Song, 2023), "The Elephant Whisperers" (Oscar, documentary short, 2023), and "Slumdog Millionaire" (A. R. Rahman, 2009 Oscars, though a British film). Institutions: Central Board of Film Certification (CBFC), Film and Television Institute of India (FTII, Pune), National Film Awards, IFFI (Goa).

## Examples and comparisons

- **Nagara vs Dravida temples:** Nagara has a curved shikhara over the sanctum and usually no boundary wall; Dravida has a stepped pyramid vimana, a boundary wall and tall gopurams at the gates, often larger than the main tower.
- **Gandhara vs Mathura sculpture:** Gandhara is Greek-influenced, grey stone, wavy hair and realistic folds; Mathura is indigenous, red sandstone, smiling face and a more robust body.
- **Kailasa Temple, Ellora:** an excavation of about 200,000 tonnes of rock, carved from the top downwards, which needed precise planning without later correction, an example of ancient engineering skill.

## Key points to remember

- Three temple styles: Nagara (Konark, Khajuraho), Dravida (Thanjavur, Madurai), Vesara (Belur, Halebidu).
- Ajanta (paintings), Ellora (Kailasa Temple), Bhimbetka (prehistoric paintings).
- Gandhara, Mathura, Amaravati schools; Chola bronze Nataraja.
- Eight classical dances with states; Natya Shastra by Bharata; nine rasas.
- Hindustani vs Carnatic; Tansen; Tyagaraja, Dikshitar, Syama Sastri.
- Phalke (1913), Alam Ara (1931), Satyajit Ray (Pather Panchali, 1955), Naatu Naatu (Oscar 2023).

## ⭐ Important questions

1. Describe the development of Indian architecture from the Harappan period to the Mughal period. Compare Nagara, Dravida and Vesara temple styles.
2. Write notes on Indian sculpture (Gandhara, Mathura, Amaravati, Chola bronzes) and painting (Ajanta, miniature and folk painting).
3. Describe the classical dances of India and the traditions of Indian music.
4. Write notes on: puppetry, folk theatre, martial arts of India, fairs and festivals.
5. Discuss India's cultural contributions to the world.
6. Trace the history and growth of Indian cinema.

# Last-hour checklist

☐ Four theories of the state and their Indian sources; matsya nyaya.

☐ Saptanga and mandala theories; Vedic tribe to Mauryan Empire; council of ministers.

☐ Purusharthas, varna, jati, ashramas, eight forms of marriage, position of women, slavery in the Arthashastra.

☐ Harappan and Brahmi scripts; four Vedas, Vedangas, Upanishads; epics and Puranas.

☐ Tripitaka, Jain Agamas, Arthashastra, Kalidasa and other Sanskrit authors.

☐ Sangam works; Telugu, Kannada and Malayalam writers; Persian, Urdu and Hindi writers.

☐ Harappan and Vedic religion; Buddhism (Four Truths, Eightfold Path); Jainism (triratna, five vows).

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☐ Six darshanas table; Shankara's Advaita and four mathas; Charvaka and Ajivika.

☐ Bhakti saints (Alvars, Nayanars, Kabir, Nanak, Tulsidas, Mirabai) and Sufi orders (Chishti).

☐ Reform movements table.

☐ Aryabhata, Brahmagupta, Bhaskara II, Madhava; Charaka and Sushruta; Kanada.

☐ Iron Pillar, wootz steel, zinc; Harappan town planning; water management examples; textiles; trade routes; GDP share.

☐ Temple styles; rock-cut caves; Indo-Islamic monuments; sculpture schools; coins; painting.

☐ Classical dances table; Hindustani vs Carnatic; theatre; puppetry; martial arts; festivals; cinema milestones.

## Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

| # | Title | Author(s) | Publisher | Where to get it legally |
|---|---|---|---|---|
| 1 | Cultural Heritage of India: Course Material | V. Sivaramakrishna (Ed.) | Bharatiya Vidya Bhavan | WorldCat |
| 2 | Indian Art and Culture | S. Baliyan | Oxford University Press | WorldCat |
| 3 | Modern Physics and Vedanta | Swami Jitatmananda | Bharatiya Vidya Bhavan | WorldCat |
| 4 | Readings in Early Indian History | Romila Thapar | Oxford University Press | WorldCat |
| 5 | The Tao of Physics | Fritjof Capra | Shambhala | WorldCat |
| 6 | The Web of Life | Fritjof Capra | Anchor Books | WorldCat |
| 7 | Tarkasangraha of Annambhatta (translation) | V. N. Jha | Chinmaya International Foundation | WorldCat |
| 8 | Yoga Sutras of Patanjali | (translation) | Ramakrishna Mission | WorldCat |
| 9 | Yoga-darshanam with Vyasa Bhashya | G. N. Jha (translator) | Vidyanidhi Prakashan | WorldCat |
| 10 | Science of Consciousness, Psychotherapy and Yoga Practices | R. N. Jha | Vidyanidhi Prakashan | WorldCat |
| 11 | Shodashang Hridayam (translation) | P. R. Sharma | Not given in the syllabus | WorldCat |
| 12 | The Wonder That Was India | A. L. Basham | Rupa & Co. / Picador | WorldCat |
| 13 | Aspects of Political Ideas and Institutions in Ancient India | R. S. Sharma | Motilal Banarsidass | WorldCat |

The syllabus lists Capra's "The Wave of Life"; his book with the closest title is "The Web of Life", listed above. NCERT's free textbooks on ancient Indian history and on Indian art and culture (ncert.nic.in) are good, legal companions for Units 2 to 5.

**Syllabus source:** AKTU B.Tech 3rd year non-credit courses syllabus (copy hosted by an affiliated college, PDF). BNC602 is the current code; that syllabus names the course "Indian Tradition, Culture and Society" (KNC602), with the same five units.